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Cross-layer multipath routing scheme for wireless multimedia sensor network

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Abstract
Multimedia transmission in wireless multimedia sensor network requires restricted quality of services (QoS) conditions. Where the resource-constrained nature of wireless multimedia sensor network (WMSNs) adds more challenges. Such a situation demands a routing strategy that can ensure QoS transmission and exploit the resources efficiently. At present, the single-path routing techniques cannot guarantee the end to end QoS requirements of video transmission causing the video quality degradation. Besides, the traditional network layers stack is not suitable for dynamic wireless sensor network environment, where important information is required from nonadjacent layers to enhance the routing decision. This paper proposes the design of cross-layer multipath routing (CLMR) scheme in order to ensure QoS and minimize energy consumption. CLMR is designed to determine the suitable multipath and to send the multimedia packets according to their importance. The cross-layer design between application, network and physical layers is adapted to obtain optimal routing decision. The simulation results show that the CLMR has a significant improvement in performance as compared to other similar existing protocols, where the obtained average delay with high delivery is less than 150 ms, while the energy consumption is reduced, and the PSNR values for the tested clips are more than 30 db.

Keywords WMSNs · Multipath · Cross-layer · Routing

1 Introduction

In wireless multimedia sensor networks (WMSNs), the sensor nodes are equipped with cameras, microphones, and other sensors generating multimedia data. The revolution in multimedia sensors is the outcome of current spreads in imaging technologies and micro-electro-mechanical systems, creating small, low-power, and low-cost image/video capture devices at a large scale that maybe within direct approach in the near future [1]. Multimedia Sensors retrieve video, audio, images and physical scalar data from the physical environment. Furthermore, the sensors have the ability to store, process, retrieve and correlate multimedia data in real-time and further to aggregate data that is initiated from heterogeneous sources. The high amount of data is generated by multimedia sensors that are going to be processed and transmitted within the network, which add more requirements to WMSNs than scalar data in traditional WSN. The multimedia transmission requirements add more resource constraints, which include energy consumption, memory, buffer size, processing abilities and bandwidth.

WMSNs expands existing WSN applications and empowers a wide horizon for new applications such as, traffic managing, automated aid systems, multimedia surveillance, environmental observing and industrial control [2]. It is a real challenge to meet the requirements of QoS for multimedia transmission within the aforementioned constraints. The stated challenges, requirements and characteristics of designing WMSNs expose various research topics and future research issues in order to improve algorithms, protocols, architectures, devices and test beds and to prolong the sensor network lifetime while

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satisfying the QoS requirements of the various multimedia applications [3].

The concept of multipath gives the source choice of multipath to transfer data where these paths may have different QoS metrics. Multipath techniques in WMSNs are attracting researchers' interest due to the variety of solutions that enhance multimedia transmission in WMSNs. Hence, the traditional layer concept in which every layer can communicate with adjacent layers is not adequate for wireless sensor network due to the unreliable, dynamic and error prone channels. Furthermore, the layered protocol stack requires high processing between layers and considerable memory size which contrasts with resource constraint nature of WMSNs [6]. In addition, the event-centric methods of WMSNs impose application aware communication protocols. These aforementioned features have motivated cross-layer protocols that share the information of two or more layers in an integrated design.

The existing proposed multipath routing protocols still have many limitations such as the strict QoS requirement for multimedia transmission are ignored and the multipath are designed at the expense of increasing overhead. The main contribution of this paper is to propose multipath routing scheme based on cross layer design. The disjointed nodes paths are maximized to eliminate the congestion and the point of failure at the joint nodes. When paths are discovered, the overhead of the paths construction is decreased by reducing the carried data. The multimedia packets are prioritized and sent according to their importance through the found paths. Heuristic cost function is designed with the most important metrics for multimedia transmission in WMSNs.

The rest of this paper is structured as follows. Section 2 provides an overview of the related works considering the cross-layer design and multipath routing in WMSNs. The proposed CLMR design is described in Sect. 3 which includes neighbor management and routing management. CLMR rerouting mechanism is discussed in Sect. 4. The simulation setup and results are discussed in Sect. 5. The future works are explained in Sect. 6. Finally, the paper is concluded in Sect. 7.

2 Related works

The dense deployment of sensor nodes can be exploited to make multipath routing an effective solution and promising technique [4]. The proposed multipath technique is different from one solution to the other, where there are some works that use one path for data transmission while using the other found paths as a backup in the cases of failure nodes. Some protocols transmit data alternatively or simultaneously in the found paths. Figure 1 shows the types of multipath routing.

In traditional networks, there are many proposed works using multipath techniques to deal with different situations like reliability, fault tolerance, congestion management, and QoS providing. However, these solutions cannot be applied directly to WMSNs. The advantages of multipath routing include reliability, security, energy efficiency, fault tolerance, bandwidth aggregation and QoS improvement [5]. However, designing of multipath techniques has many aspects to be addressed and clarified, such as the optimal number of required paths, multipath discovery, traffic allocation, resources consumption and paths maintenance. In addition, there are many issues that need more analysis and justification, like, the tradeoff between multipath and reliability, lifetime, and delivery ratio. Figure 2 shows the classification of current multipath routing protocols.

On the other hand contemporary research on WMSNs shows that cross-layer techniques achieve a considerable improvement in terms of energy consumption [7]. Cross-layer design for multimedia sensor networks aims to offer rich interlayer interaction within a node to enhance the node's performance. Despite of all the benefits of cross-layer design, there are many challenges and design guidelines which are under consideration to maximize the benefits without side effects. In literature, there are many works based on cross-layer design that will be explained in the next section. These solutions are categorized according to the interaction and information that are interchanged among the physical (PHY), MAC, network, transport and application layer [8]. Figure 3 shows the information interchanged in cross-layer design approaches.

The first Multipath routing protocol was proposed by Maxemchuk [9] to spread the traffic in space instead of time as a means for load balancing and fault treating in packet switching networks. A comprehensive survey [10] explains the operation of different multipath routing according to proposed classification, where the current multipath routing protocols are classified into three main classes: (1) alternative path routing, (2) multipath routing to achieve reliability, (3) multipath routing to utilize resources efficiently. Figure 2.7 explains the taxonomy of the existing multipath routing for WMSNs. Another review study in [11] provides a survey of different multipath routing protocols proposed for WMSNs and further classified them into three categories: (1)infrastructure multipath based, (2)non-infrastructure and (3) coding based. Macit et al. [12] presents evaluation of QoS-aware single-path and multi-path routing protocols for multimedia transmission in WMSNs. The performance results show that multi-path routing provides better reliability than routing protocol based on the single path. On the other hand, multi-path routing should be designed carefully at

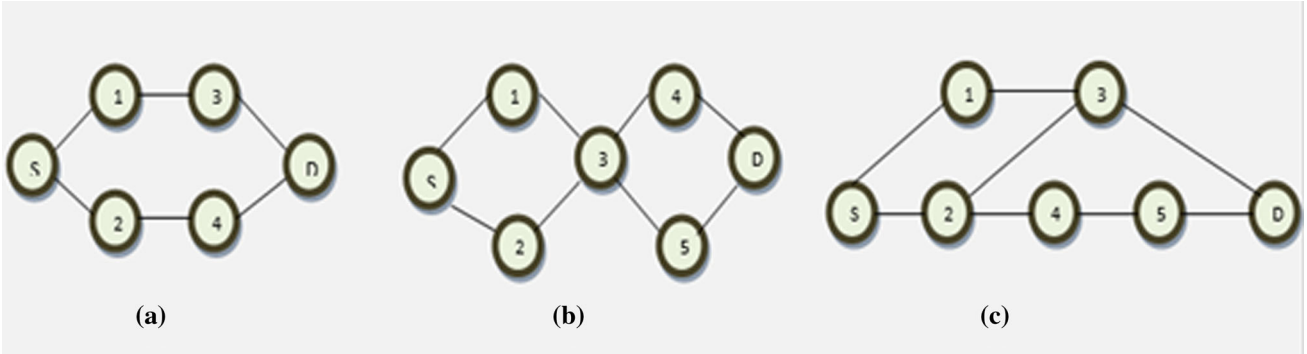

Fig. 1 Multipath routing types **a** Node disjoint path **b** Link disjoint path **c** Partially disjoint path

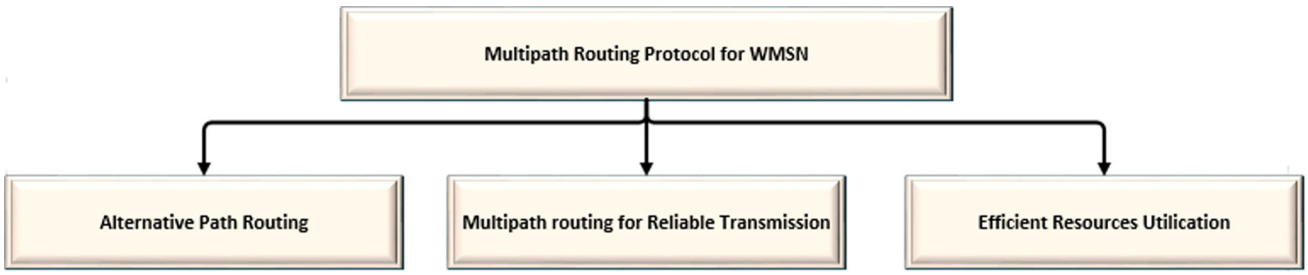

Fig. 2 Taxonomy of the existing multipath routing protocols in WMSNs

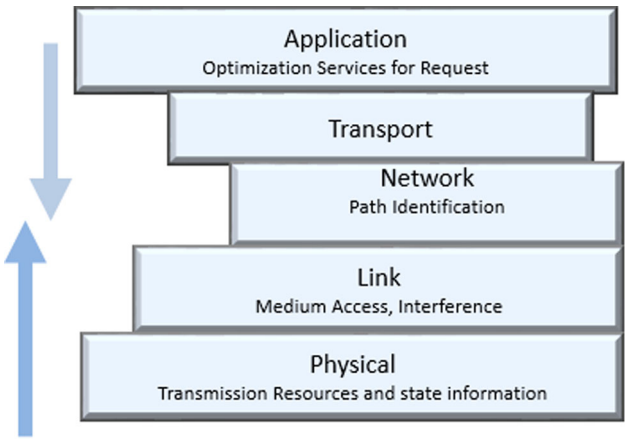

Fig. 3 The information interchanged in cross layer design approaches

high traffic in order to avoid the overhead that increases latency. Alturjman and Radwan [13] present a comprehensive overview of both best effort and real-time multipath routing protocols for WMSNs in the era of IoT. The proposed review investigates the design issues affecting the development of strategic multipath routing mechanisms for multimedia transmission in wireless multimedia sensor network.

TPGF [14] is a two-phase geographic greedy forwarding algorithm and is the first multipath routing protocol proposed for WMSNs. The first phase explores the routing paths while the second phase optimizes the found paths with the minimum number of hops between source and destination. The main problem of TPGF is only to consider the distance between the source and sink, while other important QoS metrics are ignored. TPGFPlus [15] is an extension of TPGF, which uses two-hop neighbor information for geographic routing and duty cycle assignment. The selection of the next hop is the closest node to the destination among its entire one hop and two hop neighbors. MPMPS [16] is the other extension of TPGF to prove that every path found by TPGF is not suitable for video transmission, because the long path with high delay may not satisfy the time requirements for multimedia transmission. Furthermore, the protocol distinguishes between image and audio stream of the video. The priority is given to one more than the other, according to the application requirements, to guarantee the usage of limited bandwidth and energy.

Dai et al. [17] proposed a correlation-aware QoS routing algorithm (CAQR) to transmit multimedia with QoS requirements by exploiting the correlation of visual information observed by different camera sensors. The proposed scheme provides load balancing and reduces network congestion based on the correlation of the data, which eliminates data transmission. The drawback of the proposed work is that, the correlation of the visual information needs efficient compression techniques for efficient video transmission. Sung-Lee et al. proposed an energy efficient QoS aware routing (EE-QoSR) technique for multimedia

transmission [18]. The technique is sensitive to the delay and reliability alteration even at the resource shortage. The control messages are reduced and instead, the broadcast message from the sink is utilized for reducing energy consumption.

XLP [19] is a cross-layer design based on initiative determination concept where the node decides to participate in routing based on specified rules. The node can participate in the routing, under the conditions, such as channel quality, local congestion and buffer length and remaining energy are applied. The functionalities of all layers from physical to transport are integrated into cross-layer protocol that aims to achieve congestion control, routing and medium access control. The potential receivers that meet the defined conditions are contented for packets receiving to become next hope. The congested nodes are not allowed to participate in communication to control the congested situation. The holes and local minimum are avoided by using angle-based routing. XLCU [20] is a cross-layer based protocol aims to achieve reliability, flexibility and QoS requirements in WMSNs. The functions of physical, MAC and network layers are controlled as one module, where the communication model is based on TH-IR-UWB transmission technique. A geographical forwarding mechanism is introduced according to the application requirements and hop-by-hop forwarding based on admission control. The next hop is selected according to the locally determined criteria while, the receiver acts according to local measurements, control recovery and rate adaption. The admission control is guaranteed through packet service differentiation at each hop in term of throughput, delay, reliability and energy consumption. Dynamic channel coding is introduced to adapt the interference level of receiver node. Farooq et al. [21] proposes cross-layer architecture that supports different types of flows and classifies the traffic into six types. The proposed protocol aims to minimize interference and enhance the throughput. A shared database is proposed, where each layer stores the important information in the shared database and the several objects can reach the information stored in the database. Each application has reserved storage. The MAC layer uses TDMA to deal with different types of traffic flows where multiple applications are supported on the single node.

Muhammed Usman et al. [22] proposes a framework to provide QoS in WMSNs with a light-whit-error concealment scheme to enhance the QoE. The authors' objective is to increase throughput while covering the affected dropped video packets. In addition, the authors address the problem of maintaining the quality of the transmitted video and QoS with an increase in the number of video transmitting resource. Multipath video packet scheduling and efficient video encoding scheme is utilized to improve QoS in WMSNs along with efficient energy consumption.

Shah,G.A et al. [23] proposes cross-layer framework where the QoS requirements are connected to the operation of application, network, MAC and link layers under lying video stream realization, while the QoS required on each video stream is still conserved. The forwarding zone is divided into three different parts in order to explore multiple disjoint paths and to avoid interference. The forwarding decision is based on the relying nodes, where the hop delay, bandwidth and reliability constraints are considered. The packets are delivered under probabilistic hops count limit. Traffic classifier classifies the frame types according to their priorities. Xu yang et al. [24] proposes a low duty cycle mechanism for wireless multimedia sensor network based on Wi-Fi intelligently switched OFF and ON, according to the human detection method. The human detection method leverages the physical layer of the channel state information (CSI) provided by commercial Wi-Fi. The experimental and simulation results show that the energy consumption by applying the low duty cycle mechanism has been reduced by 40% as compared to the WMSN. Mekonnen et al. [25] investigate that the energy overhead is generated by docker-based virtualization on battery powered camera sensor nodes. Most of the time the camera nodes are in the off state and are switched on just when the activity is detected. Orchestration and management of the nodes are implemented on occurrence of an event. The authors determine the optimum video streaming length where the impact of virtualization becomes negligible, which was about 1-hour long video stream. The result shows that the energy wise Docker containers are suitable for virtualizing video surveillance applications, if the cameras wake-up and stream at least for an hour.

CMVT [26] is a cross-layer based multipath video transmission scheme, where the differentiated service is introduced with multipath routing. The video frames are marked at the application layer according to their importance, which distinguish them from routing layer and forward them in different paths. The routing discovery process consists of two mechanisms, 'greedy forwarding and rollback', where the forwarding decision is taken on the basis of distance and remaining energy, while the rollback mechanism is used to deal with the situations, like holes and dead links. The guaranteed QoS level of the path is calculated to evaluate each path, and then the source node chooses the suitable path for each frame type. Islam T. Almalkawi [27] proposes CMRP cross-layer routing protocol to exploit MAC layer QoS based scheduling and to improve routing efficiency. The clusters are formed based on the received strength of signals, while the multipath is enhanced with adaptive scheduling for various traffic types. The proposed scheduling technique is built on TDMA at

two levels: first, clusters within the network and second, between cluster heads. Adaptive time slots are used while the cluster head is in charge to plan the traffic toward the destination based on the data type and its availability. The obtained information results of exchange between the neighbor nodes specify the routes to the sink node. LQEAR [28] is an energy-aware and reliable routing technique, aims to find energy-aware good quality links along the path. The criteria for selecting the next node is based on the distance, link quality and remaining energy. The link quality takes the main place in the routing decision where the link quality estimation is based on the rate of energy detection of packet reception. Simulation results show that the proposed technique offers better Packet Delivery Ratio (PDR) and balances energy consumption of nodes to extend lifetime network.

The simulation results of CLMR are compared to CMRP, CMVT and LQEAR, since all of them are multipath routing schemes specified for multimedia transmission in WMSNs. In addition, all of them follow cross-layer design with main objective to satisfy QoS requirements and to reduce energy consumption.

3 Design of CLMR in WMSNs

Designing multipath routing protocol is a challenging task, as many issues and questions are to be clarified. First, the following questions are answered that start by the way of routes or paths discovery process are answered. How does the path discovery trigger (reactive or proactive)? What is the type of multipath (joint nodes, disjoint nodes or link disjoint)? And further, how to use the discovered path in multimedia transmission? Furthermore, how does the maintenance strategy work in the case of the path failure?

The CLMR maximizes the disjoint node path and restricts the acceptance of joint nodes according to predefined conditions such as, alternative path with best optimal transmission value. In the proposed scheme, a heuristic algorithm is used based on the cost function to select the best neighbor node in terms of delay, remaining energy and ratio of packet reception. These metrics are combined into the cost function to decide the best next neighbors that build the paths to the destination. In addition, these metrics are combined with parameter weight to control their importance. The most suitable values of these parameters are obtained by using an exhaustive search optimization method. All trails are analyzed for different topologies, density and traffic load, and further the average performance is determined accordingly. The packet classifier in the CLMR classifies the multimedia packets into three packets based on data frame, which are I, P and B packets. So, at least three paths are needed from the source node to sink node, these three paths are not necessary to be disjoint node paths, rather, some paths may have joint nodes, according to the acceptance rules. When the joint nodes receive two packets, the forwarding priority will be given to the higher priority packet.

The process starts when there are multimedia packets for transmission, so the discovery process starts to collect the neighbor information and build neighbor table. After that, the best three next neighbors are selected by the source node to start multipath discovery. The second hop and intermediate nodes will send route control packet RCP to the only best neighbor until reaching the destination. When different RCP packets reach the destination from different paths, the destination will calculate the cost of each path and send the information back to the source node by using route control reply packets RCRP. The source node starts data transmission by using the selected paths according to the importance of multimedia packets as shown in Fig. 4.

3.1 Neighbor management

The nodes use the link cost function to select the best neighbor nodes during the discovery phase. The cost function takes into account delay, packet reception ratio and remaining energy. The collected metrics for neighbor nodes are combined into the cost function to define the best neighbor nodes that can participate in the path establishing from source to destination. After that the route control packets RCP are sent to the best neighbors and the RCP packet carries the cost of each link. The neighborhood management is designed in CLMR to discover neighbor nodes Ni and saves them in the neighbor table to be ready for data transmission. Due to the memory constraint, the neighbor table is limited to a specific number of nodes that are mostly applied in the optimal transmission requirements.

3.1.1 Neighbor discovery

Initially, the neighbor discovery phase is started at each node that will be a part of multipath by broadcasting hello packets. Each neighbor node that receives hello packet will be replied by sending information, remaining energy, delay and the packet reception ratio. After that, each node calculates the cost of neighbor nodes based on the received metrics to select the best nodes as shown in Fig. 5. If the source node S does not get any replay within a certain period of time, the rerouting mechanism is invoked. The optimal Transmission function is proposed to calculate the cost for each neighbor. The best next hop node which is determined by the optimal transmission values i.e.$Best \rightarrow OT(t)$ will be used to discover the full paths. The considered metrics are chosen to enhance the network

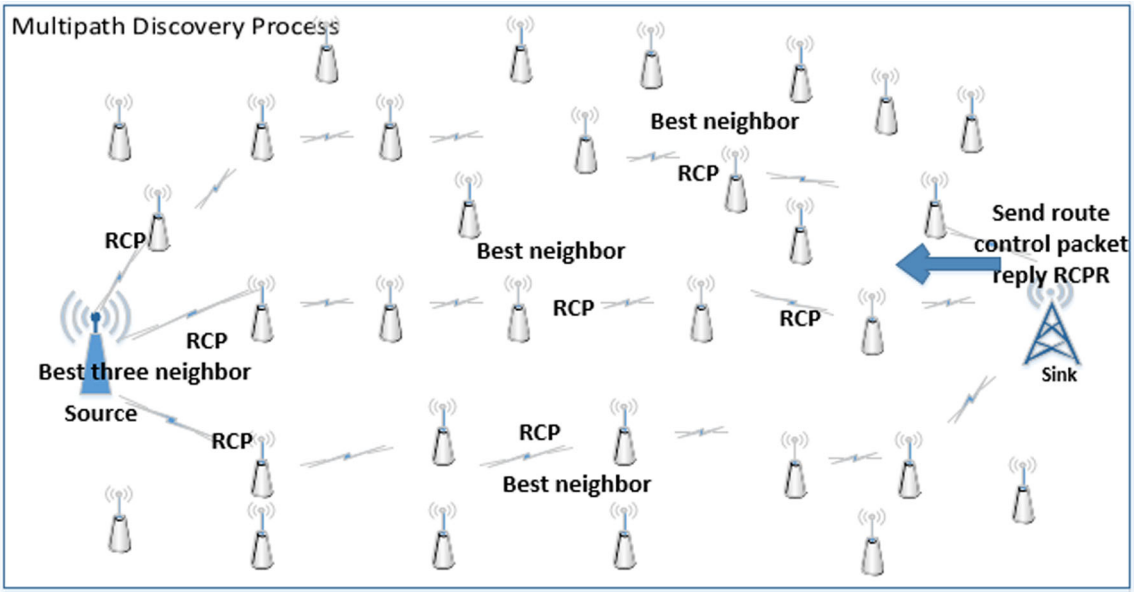

Fig. 4 Multipath discovery process

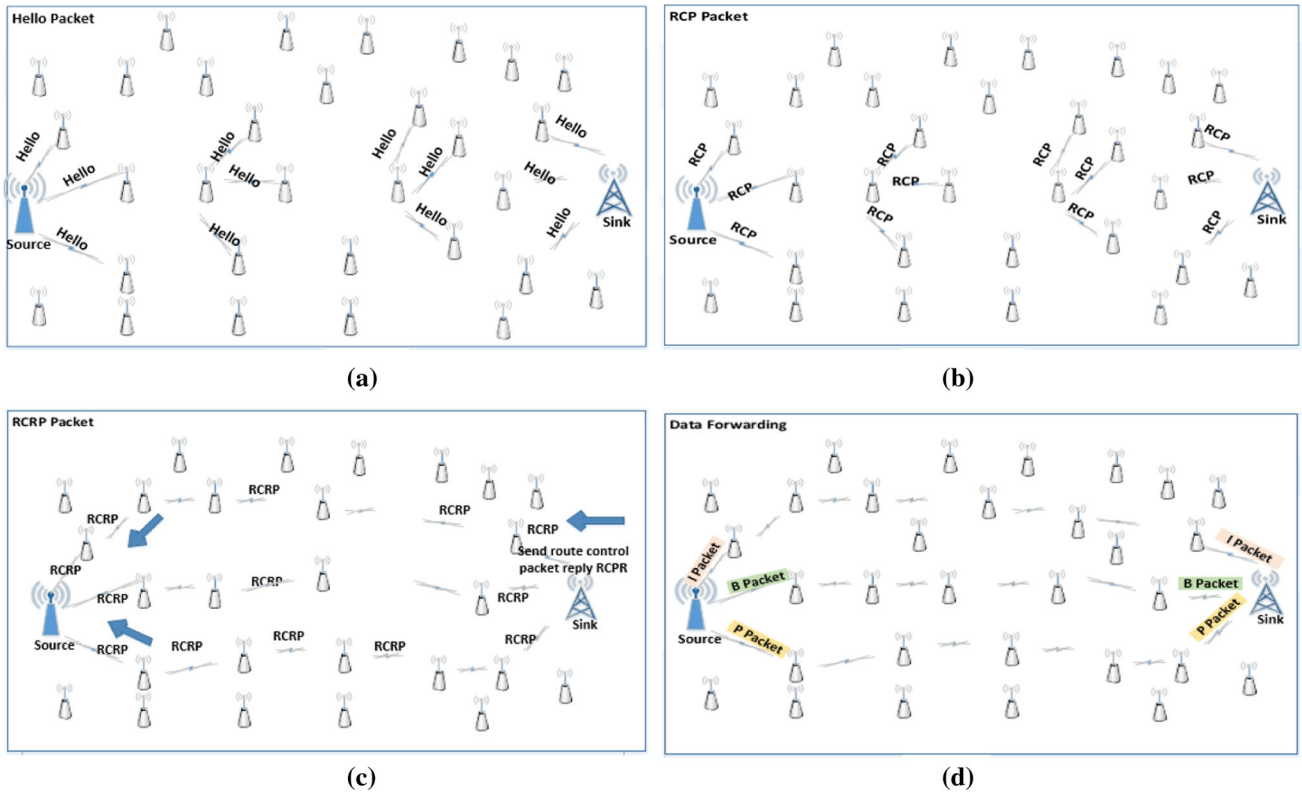

Fig. 5 Multipath discovery phases **a** Hello packet exchange **b** Route control packet **c** Route control reply packet retransmission **d** Data forwarding

performance, where the delay plays main role to improve the multimedia transmission, while the packet reception rate reflects the link quality and enhances throughput and reliability. Energy metric is very important to prolong network lifetime within the limited energy resources.

3.1.2 Optimal next hop decision

The optimal cost function is used to calculate the best neighbor nodes taking into account the QoS requirements

of WMSNs, so that the delay, packet reception ratio and energy maybe considered.

The optimal transmission from current to next node is mathematically expressed as follows:

$$OT = MAX\left\{(\omega \times P_{rr}) + \beta \times \left(\frac{E_c}{E_{max}}\right) + \gamma \times \left(1 - \frac{D}{D_{max}}\right)\right\}$$

where P_{rr} is a percentage value that reflects the link quality, E_c is the current energy value, E_{max} is the maximum battery voltage for sensor nodes. D is the average one hop delay, D_{max} is the maximum end-to-end delay for multimedia application, which is 250 ms. ω, β and γ are parameters weights that control the importance of each metric for specific application needs. These parameter values are estimated by an exhaustive search using Network Simulator-2 (NS-2) in the way: $\omega + \beta + \gamma = 1$.

3.2 Routing management

In the routing management, the multipath is discovered from source node to destination; as a result, the nodes should have the routing tables that add flexibility and more capabilities. The nodes that construct the path to the destination are saved in the routing table along with their optimal transmission value. Having finished the neighbor discovery phase, the nodes will have information about their neighbors and every node will calculate the optimal transmission value for each neighbor and choose the best neighbors by sending the route control packet RCP to discover the full path. The RCP moves from node to the next on the way to the destination node and it carries the optimal transmission value for each passed link while moving. CLMR strategy is to decrease the carried information leading to save the limited resources of sensor nodes during multi-hop transmission. When the RCP reaches the destination node from different paths, the sink node will check the total optimal transmission value for each path *OP* and sends back rout control reply packet RCRP by using the same nodes that construct the multipath. The RCRP carries the total optimal transmission value for each path and goes back hop by hop until it reaches the original source node. When the source node gets RCRP from different paths, the multimedia packets are sent through the suitable path. Routing management module is also responsible for nodes failure, so in such case, a proper action will be taken by starting the process of rerouting.

3.2.1 Multipath discovery

The source node starts a multipath discovery phase by sending RCP (Route control Packet) to the best three neighbor nodes in the neighbor table which have high optimal transmission values, and they are known by the neighbor discovery phase. Every packet has a source ID and a unique generated number. In the next step, every node receives RCP packet and it will act as follows: First check, if the packet is received for the first time, then it will be added with the optimal transmission value *OT* and will be forwarded to the best neighbor node in the way to the sink node. There are three scenarios to accept or discard the packet, in case the RCP is received already, as shown in Fig. 6. First, if the total optimal transmission value *OT* of the packet is better than the received one, the packet will be accepted in any way. In the other case, if the *OT* value of the RCP is less than the received packet, there are two different cases. If the packet is subset from the received one, then it will be dropped directly. The second case, when both the received packets are not overlapped each other; an acceptance factor ε is introduced ranging from 1 to 2. If the *OTvalue* of RCP2 is less than that of *OTvalue* of RCP1 (the received one), it will be accepted if $RCP2 < RCP1/\varepsilon$. Where ε is the factor for a node to take a decision whether to accept or to reject a particular RCP packet that has less total *OTvalue* than the packet, which has come before the same destination. This means that if a packet has less *OTvalue* than the arrived one, it will be accepted. This factor can be changed according to the network requirements. In CLMR, after experimenting with different values, the value of $\varepsilon = 1.5$, when ε is kept with values bigger than 1.5, the obtained *OTvalue* leads to finding out paths that degrade the overall performance. The aim of dropping the repeated packets is to maximize disjoint multipath from source to destination. The nodes that hop away from the sink node are allowed to accept more than RCP, because the nodes near to sink node get more RCP packet than other nodes. When the first packet reaches the sink node, the sink will await the maximum delay time for other packets coming from different paths. This time is related to the first packet's arrival time and aimed to eliminate the bad paths. The time is calculated as follows:

$$W_t = 250 - D_{t(i)}$$

250 ms is the maximum delay, and should be less than or equal to 250 ms for multimedia applications. $D_{t(i)}$ is the total delay of the first packet reaches from path i. The packets reached after this time will be considered as bad paths and will be excluded from multipath calculation. When different RCP packets are received from different paths, the sink node will calculate the *OP* value for each path individually; it will be explained in the next section.

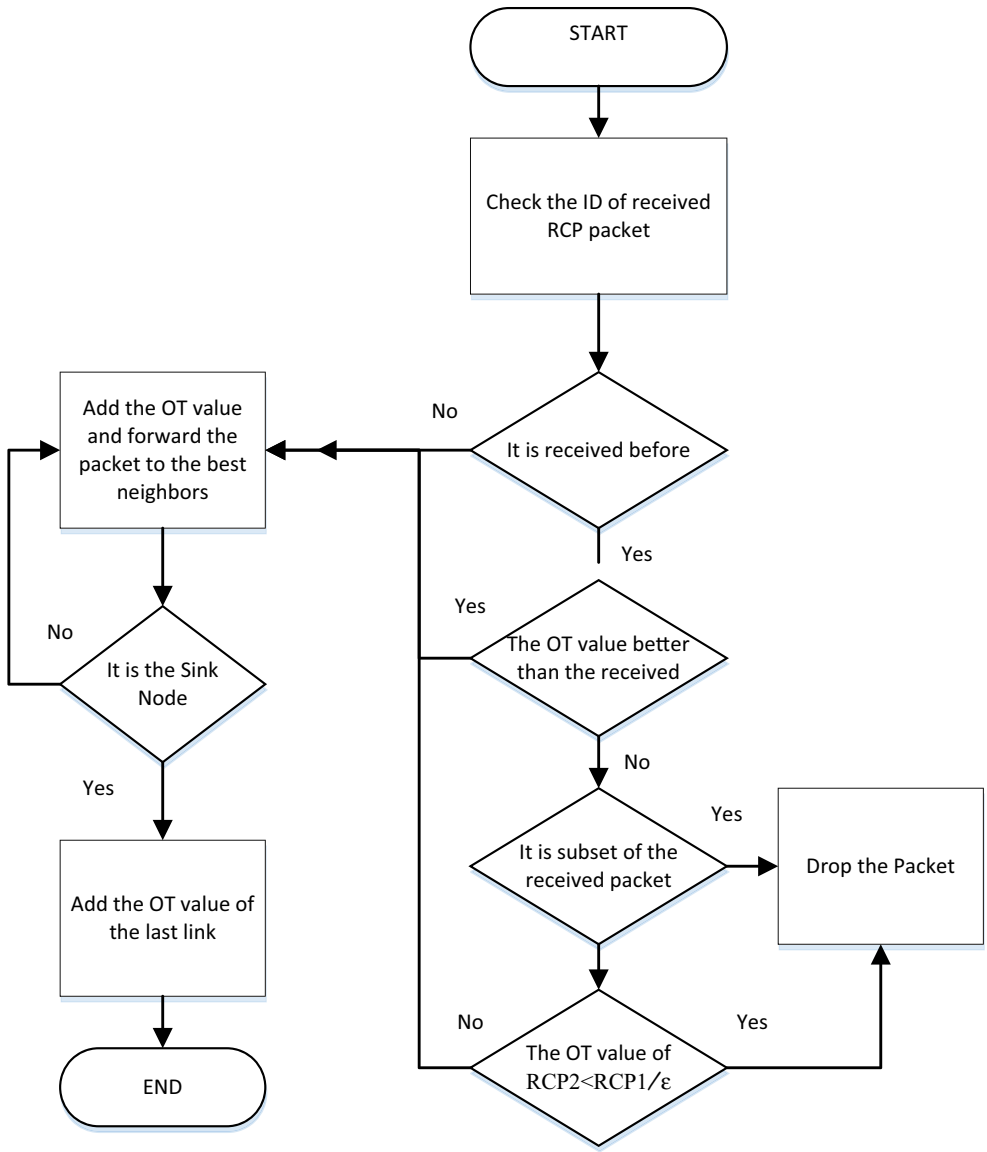

Fig. 6 The process of sending and accepting RCP

3.2.2 Optimal multipath decision

As mentioned before, the RCP packets reach the destination node by carrying the accumulated *OTvalues* of each found path. The sink node receives the cost of all paths defined by the total OT. Let N_i be the nodes, which construct the Path p_i from source to destination. The cost value of each path is defined by $OP.OP$ is calculated as follows:

$$OP_j = \sum_{i=1}^{n} OT_i$$

Where j is integer value $1, ..M$

When the costs of all found paths are calculated at the sink node, the paths cost will be sent back to the source node, so the source node has information about the found paths to the destination. The paths' costs will be arranged in order, from OP_j $toOP_M$. When the data transmission starts, the source node will select the best paths according to importance of packet, and it will be explained in the next section.

3.2.3 Data forwarding

Concentrating on the volume of the compressed data, the video frames are compressed by using different algorithms in the domain of video compression. These video frames are called picture types or frame types [29, 30]. These frame types are $I, PandB$ where they are different in their features and importance.

After the selection of a set of multipath, the source node starts sending video frames to the destination along the found paths. The application layer sets the frame type into the header of each packet by appending an extra bit of information to act as a stamp and to distinguish between frame types, which can be recognized by the routing module. The packet classifier is used in CLMR to support different packet types, according to their importance, and to classify the received packets based on data frame into three priority packets, which are $I, PandB$ packets, as shown in Fig. 7. Since the classifier set three types of packets, a priority queue is used to route the multimedia packets based on the packet priority. In a priority queue, a packet with high priority is served before a packet with low priority. If two received packets have the same priority, they are served according to their order in the queue [31] [32]. So I frame will be sent by using the best found path, P frame uses the second-best path and the third best path is specified for B frame. This distributing mechanism improves the reliability and the load balancing in CLMR. Moreover, when the joint nodes between paths receive more than one type of packet, the high priority packet is sent through the best path and further it sends feedback control packet to the node that forwards the less priority packet. When the feedback control packet is received, the node will change the forwarding to the next optimal node. For example, in case the joint node receives I-packet which interferes with B-packet, the joint node will forward both packets and send a feedback control packet to the node that sends B-packet. The node that sends B-packet will select another path; in addition, I-packet path interferes with P-packet path. CLMR at a joint node will forward both packets and sends feedback control packet to the node that sends P-packet. The node that sends P-packet will select another path. The third scenario is that I-packet path interferes with P-packet and B-packet paths. CLMR protocol at joint node will forward all packets and sends feedback control packet to the nodes that send P-packet and B-packet. The nodes that send P-packet and B-packet will select another path. It is the last case when P-packet path interferes with B-packet path. CLMR at joint node will forward both packets and sends feedback control packet to the node that sends B-packet. The node that sends B-packet will select another path. In the next search process, the optimal nodes are selected based on pre-stored optimal transmission values OT. The nodes that are neighbors to the sink node forward the data packet directly. In case of failing to find the next hop node, or when there is no neighbor node in the routing table, the rerouting phase starts to recover the failure nodes.

4 Rerouting

The CLMR policy is to maximize the number of disjoint paths by restricting the acceptance of RCP based on the aforetime rules. Due to this policy and the dynamic nature of sensor nodes, some nodes may go in block situation. The blocked nodes are the nodes that do not have next neighbor node in the way to the destination node. Furthermore, the blocked nodes can happen due to the energy depletion of the next node. While the data transmission starts in the selected joint or disjoint nodes path, the intermediate nodes keep the routes situation checking. In case the node loses connection with its neighbor (do not receive live packets from its neighbor) this means that the next node becomes unreachable and becomes blocked node. The maintenance process goes through two phases. The first one is called rapid intervention, whereas the transmission power is increased to find further neighbor node as explained in

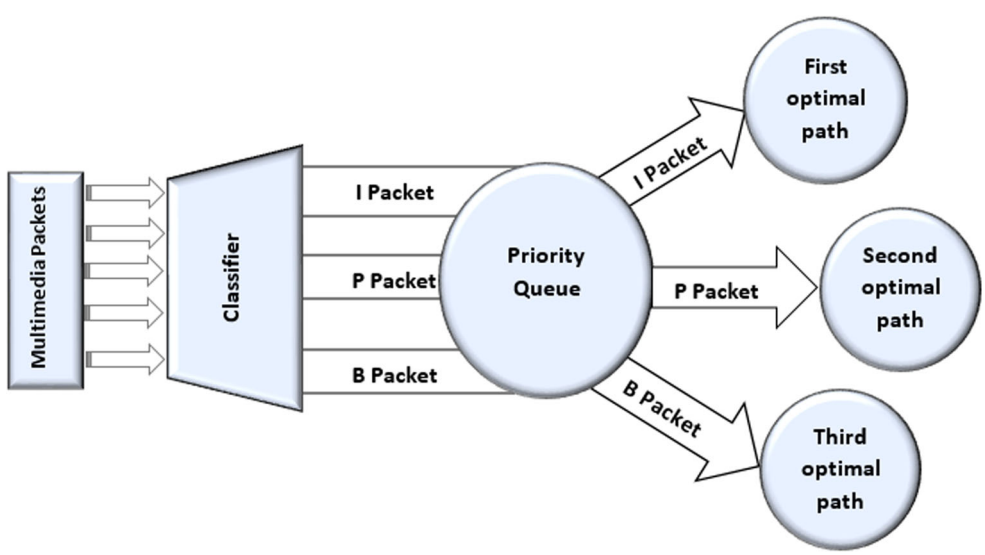

Fig. 7 Packet Classifier and priority Queue in CLMR

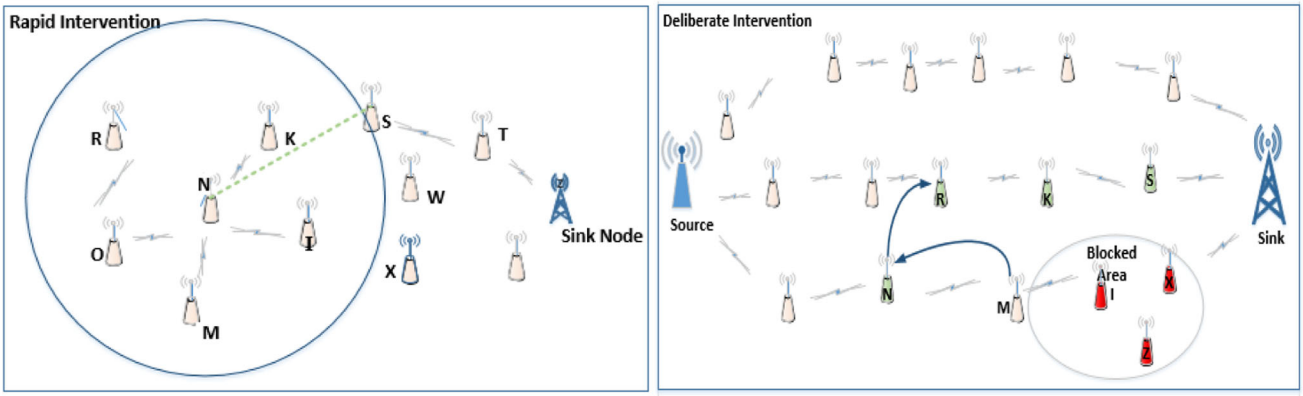

Fig. 8 Rerouting mechanism (rapid and deliberate intervention)

Fig. 8. The source node N looses the connection with its neighbors $K, R, Q, O, MandI$, so in this case the blocked node will use maximum power transmission to connect with node S. If the rapid intervention fails to find the next neighbor node, deliberate intervention is applied where the blocked node sends block node packet to its parent node, so it will stop sending to this node and use the second optimal node in the neighbor table, as explained in the Fig. 8. The rerouting management is also invoked, in case the energy of specific node becomes less than defined threshold. The parent node will be informed about the energy situation and accordingly the node will be excluded from the next route selection. The repaid and deliberate intervention strategies improve the performance of the proposed scheme, by decreasing the overhead and the time of discovering and establishing new paths, as a result, the energy consumption will be reduced and prolong network lifetime.

5 CLMR performance evaluation

NS-2 simulator has been used to simulate the CLMR routing scheme. IEEE 802.15.4 MAC and physical layers are used to reflect the real mechanism access in WMSNs. The CLMR performance is analyzed and compared with LQEAR, CMRP and CMVT protocols. CLMR routing scheme has been simulated, based on the characteristics of the LOTUS mote from MEMSIC [33]. The traffic is typical among multiple source nodes sent to the base station. A grid topology is used where 121 nodes are deployed in the area (110 m x 110 m). Evalvid application is used at application layer with optimal parameters. CLMR is applied as a routing protocol in the network layer. CLMR has been tested on three real-time video clips from the website which are, Akiyo, Foreman and highway as explained in Fig. 9. In the evaluation study, the metrics of delivery ratio, delay, energy consumption, packet overhead, PSNR, jitter and packet loss rate are considered. Table 1 displays the simulation parameters that are used in the simulation experiments.

The performance of delivery ratio, achieved by CLMR is higher up to 18,14 and 10% as compared to CMVT, CMRP and LQEAR respectively, as shown in Fig. 10a. CLMR uses the high-quality paths for data transmission where the priority is given to I packets to be sent through the best path which has less delay and packet reception rate obtained from the cost function. In addition, CLMR maximizes the disjoint node path which reduces the nodes congestion and packets loss. The highest delivery ratio in CMVT is reached up to 80% where the packets are sent according to their importance, but as mentioned the links quality is not considered where the routing decision depends on the number of packets sent by each node on the path beside the distance and hop count. CMRP uses two clustering levels which increase overhead at cluster heads leading to congestive situation. LQEAR has the highest delivery ratio after CLMR where the link quality and packet reception rate are considered to calculate the cost function. However, there is no consideration for the disjoint node paths while joint nodes increase the probability of congestion especially near to the destination node. Figure 10b shows the loss rate of all packets. The figure shows that the loss packets of the most important packets like *IandP* are less than 0.0017 and 0.015 respectively in the worst case in CLMR. The previous explanations are applied here as well, where the joint nodes used in LQEAR, CMVT, and CMRP rise the probability of nodes congestion and its consequences. The packet loss rate is changed with different video clips based on the video motion and complexity.

The end-to-end delay result is presented in Fig. 11a. It is clear that the CLMR delay is 11% less than LQEAR, 15% than CMRP and 19% than CMVT. In the maximum cases, the delay does not exceed the set limit by 250 ms.

In CLMR high-quality links are used to send the multimedia packets especially the packets that belong to I

Fig. 9 Snapshots of received video clips

Table 1 Simulation parameters used in the simulation experiments

Propagation model	Shadowing
PhyType	Phy/WirelessPhy/802_15_4
MacType	Mac/802_15_4
freq_	2.4e + 9
Initial energy	3.6 Joule
Transmission power	1 mW
Traffic	Evalvid application and VBR
CODEC	H.264
Frame rate	20 fps
Source width	176 Pixel
Source height	144 Pixel
Number of frames per group	15

frame. In addition, the congested nodes are avoided by maximizing the disjoint path and giving the priority to the important packet at joint nodes. In CMRP, the cluster heads aggregate packets from different nodes and after that they are forwarded to the destination node. This arise issues of point failure besides the collision and congestion at cluster heads, where there are two cluster levels. LQEAR and CMVT use distance metric as indicator of delay which is not necessary, as the short distance links have less delay. Figure 11b shows the average jitter values for all the clips by using CLMR, LQEAR, CMVT and CMRP protocols. The results show that the jitter values for CLMR are within the limited range. The jitter values are ranged from 0.0001 s to 0.00025 s, the clips with high motion have more jitter than other videos. CLMR exploits the high-quality links for data transmission where the packet reception rate is calculated in the cost function. In addition, CLMR uses the rapid intervention in case of nodes failure, so most links are stable with less delay and high quality. There is no clear mechanism in LQEAR, CMVT, and CMRP protocols to recover the failed nodes on the way to the destination.

The obtained PSNR values of CLMR for all clips have acceptable PSNR values, which are greater than 30 db that means it is good in MOS scale. The results of PSNR values are shown in the Fig. 12a. As it has been explained already the importance of P and B frames is less than I frame in video construction. Accordingly, the loss of P and B frames do not affect the video quality as compared to I frame, so in CLMR more attention is given to I frame by sending all packets belonging to I frame through the best path. The quality of all the received videos is within an acceptable range. The PSNR values of CLMR outperform the LQEAR, CMVT and CMRP by 8, 12 and 10% respectively. CMVT and CMRP have convergent values, where the quality is considered in both of them, following different approaches. The clusters are formed in CMRP base on RSSI values. The CMVT considers the quality as a measure of transmitted packets through each path and differentiates between video packets by sending packets belonging to I frame through the best-found path. Furthermore, the probability of error was varied from 0 to 0.3 at the interval of 0.05 in order to check the channel effects. Figure 12b shows that the video quality is reduced with an increase in the probability of error. In addition, it can be

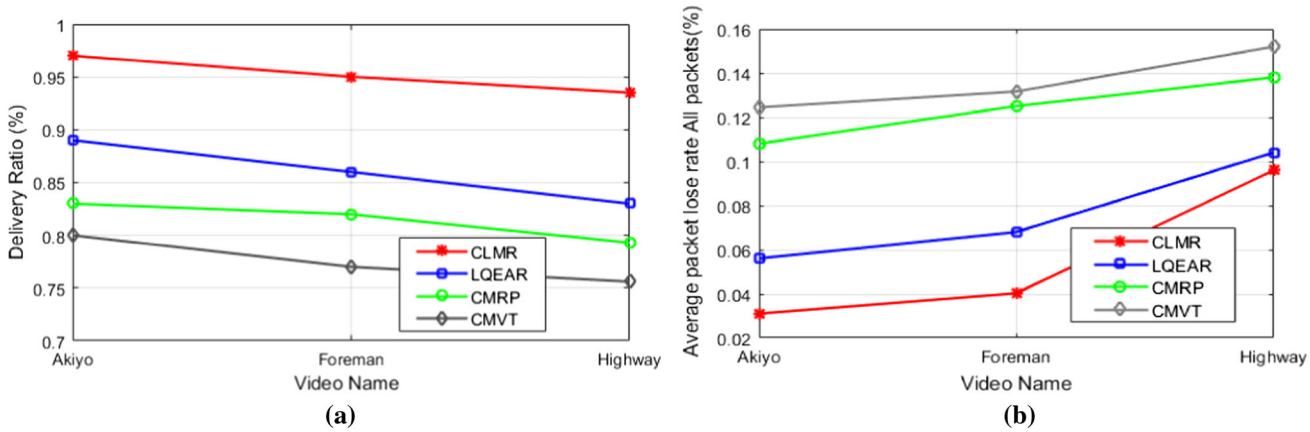

Fig. 10 Simulation results comparison. **a** Delivery ratio, **b** Average packet loss

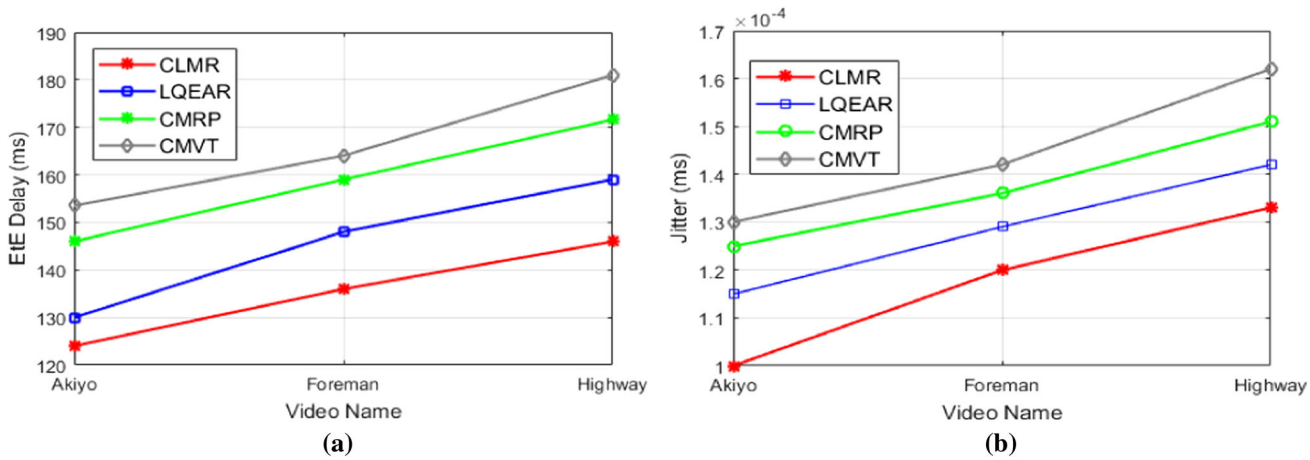

Fig. 11 Simulation results comparison**. a** End-to-End, delay, **b** Average jitter

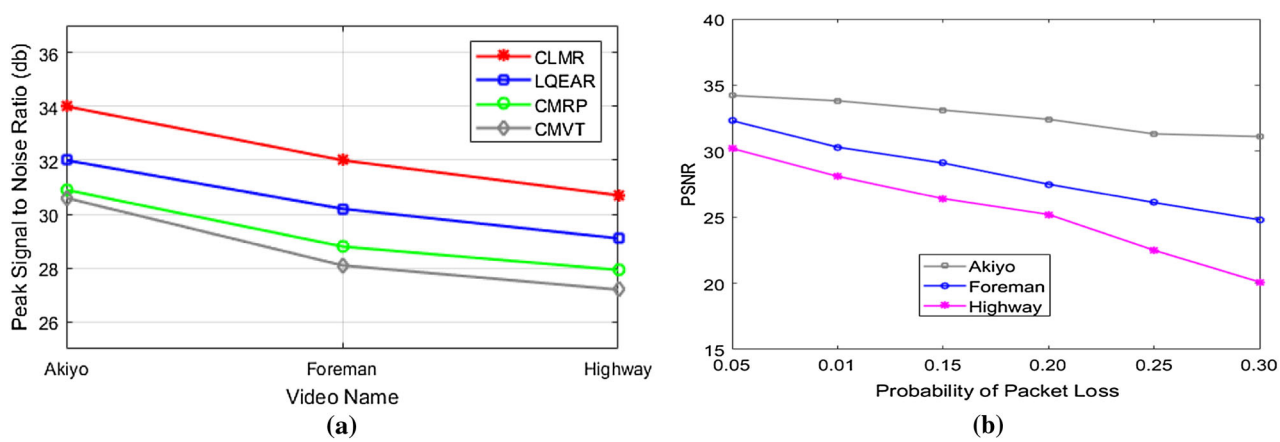

Fig. 12 **a** PSNR evaluation of transmitted clips, **b** The probability of packet loss

observed that the transmission error affects certain samples more than the others, although the test and setting condition are the same. This is primarily due to the difference in motion and scene complexity of the video sequence. The level of motion, background and camera play an important role in the nature of the video samples.

Figure 13 shows the energy consumed by the transmitted video clip; the results show that the CLMR consumes

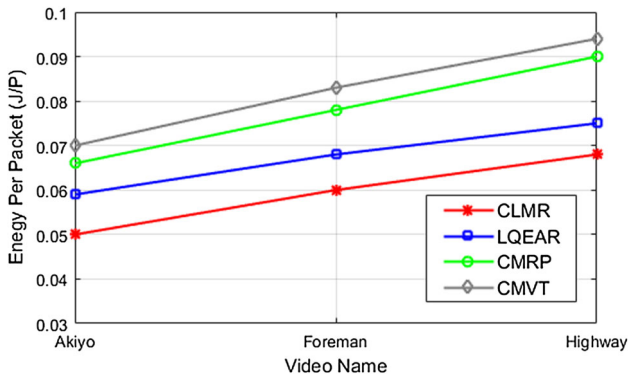

Fig. 13 Energy consumption comparisons

less energy than LQEAR by 10%, CMVT by 28% and less than 23% by CMRP. The CLMR strategy reduces the overhead and utilizes the resources by considering the energy consumption metrics. The selected high-quality disjoint node paths reduce packet-loss that lead to less packet retransmission decreasing the consumed energy. In addition, in the case of nodes failure, the rapid intervention reduces the effects of failure nodes.

6 Future work

As a future work, there are several issues that can be addressed. The current work is extendable for future suggestions:

i. Cognitive radio networks: CLMR scheme can be extended to exploit the features of cognitive radio networks. CRNs have been widely endorsed as a promising solution to mitigate the spectrum underutilization [34].
ii. Integration with the internet and other wireless technologies: The CLMR is a standalone system without internetworking to the Internet. The future work is to integrate CLMR, the Internet and the cloud network. Integration can lead to IoT allowing data to be remotely accessible via the Internet by using other networks and platforms.
iii. Energy harvesting: Recently, there has been more interest in the network made of sensor nodes without batteries and nut harvests the required energy from outside. By implementing energy harvesting techniques, potentially infinite energy becomes available to sensor nodes [35].
iv. Mobility support: In CLMR, all nodes are considered stationary nodes. The work can be extended by introducing mobility to specific node like a sink node that helps to ensure the coverage and reduces the number of nodes needed to cover a specific area.
v. Context-aware sensing in WMSNs: CLMR can be extended to be a part of an integrated sensing information. This integration will make CLMR-DDC intelligent and to have the capability to make the suitable reaction, according to the event occurred.
vi. Multipath tradeoff issues: there are trade-off issues in multipath routing that need more investigation and analysis which are not included in this work, such as the effect of frequently network topology changing, the trade-off between reliability and lifetime and the trade-off between traffic overhead and reliability.

7 Conclusion

Multimedia applications expand the usage of Wireless Sensor Network (WSNs) on different aspects and open many research challenges. In this paper a cross-layer multipath routing scheme CLMR is presented. The multipath are discovered from source to destination based on a heuristic cost function. The cross-layer design among application, network, and physical layers are adapted to obtain optimal routing decision. The metrics of delay, energy and packet reception rate are combined to select the optimal paths. The multimedia packets are classified based on their priorities and are sent through the suitable paths where the disjoint node paths are maximized. The simulation results show that the CLMR outperforms LQEAR, CMVT and CMRP protocols. The delivery ratio of CLMR reaches 95% and the obtained delay is less than 150 ms. Energy consumption, on average is less than 20% being compared to similar existing protocols, and the peak to signal noise ratio (PSNR) values are greater than 30 db which means that they have good quality. In addition, the CLMR performance proves its suitability for multimedia transmission in WMSNs.

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References

1. Charfi, Y., Wakamiya, N., & Murata, M. (2009). Challenging issues in visual sensor networks. *Wireless Communications, IEEE, 16*(2), 44–49.
2. Akyildiz, I. F., Melodia, T., & Chowdury, K. R. (2007). Wireless multimedia sensor networks: A survey. *Wireless Communications, IEEE, 14*(6), 32–39.
3. Waltenegus Dargie, C. P. (2010). Fundamentals of wireless sensor networks. In D. Y. P. Xuemin (Ed.), *Wiley Series on Wireless Communications and Mobile Computing*. United Kingdom: Wiley.

4. Lou, W., Liu, W., & Zhang, Y. (2006). Performance optimization using multipath routing in mobile ad hoc and wireless sensor networks, in combinatorial optimization. In M. Cheng, Y. Li, & D.-Z. Du (Eds.), *Communication networks* (pp. 117–146). Berlin: Springer.
5. Radi, M., et al. (2012). Multipath routing in wireless sensor networks: Survey and research challenges. *Sensors, 12*(1), 650.
6. Jurdak, R., (2007) *Wireless ad hoc and sensor networks a cross-layer design perspective*. Springer series on signals and communication technology. Berlin: Springer. https://doi.org/10.1007/978-0-387-39023-9.
7. Han, G., et al. (2015). Cross-layer optimized routing in wireless sensor networks with duty cycle and energy harvesting. *Wireless Communications and Mobile Computing, 15*(16), 1957–1981.
8. Srivastava, V., & Motani, M. (2005). Cross-layer design: A survey and the road ahead. *Communications Magazine, IEEE, 43*(12), 112–119.
9. Zytoune, O., Y. Fakhri, and D. Aboutajdine. *An energy aware QoS routing protocol for wireless sensors network*. In *Multimedia computing and systems, 2009. ICMCS '09. International Conference on*. 2009.
10. Wang, H., et al. (2010). A survey on the cross-layer design for wireless multimedia sensor networks. In Y. Cai, et al. (Eds.), *Mobile wireless middleware, operating systems, and applications* (pp. 474–486). Berlin: Springer.
11. Zhuo, X., Loo, J., Cosmas, J., & Yip, A. (2008). Distributed video coding in wireless multimedia sensor network for multimedia broadcasting. *WSEAS Transaction on Communications, 7*(5), 418–427.
12. Winkler, T., & Rinner, B. (2014). Security and privacy protection in visual sensor networks: A survey. *ACM Computing Surveys, 47*(1), 1–42.
13. Al-Turjman, F., & Radwan, A. (2017). Data delivery in wireless multimedia sensor networks: Challenging and defying in the IoT Era. *IEEE Wireless Communications, 24*(5), 126–131.
14. Lei, S., et al. (2008) Geographic routing in wireless multimedia sensor networks. In *Future generation communication and networking. FGCN '08. Second International Conference on* 2008.
15. Dong, Y., et al. (2013). Two-Hop geographic multipath routing in duty-cycled wireless sensor networks. In H. Qian & K. Kang (Eds.), *Wireless internet* (pp. 155–166). Berlin: Springer.
16. Zhang, L., et al. (2008). Multi-priority multi-path selection for video streaming in wireless multimedia sensor networks. In F. Sandnes, et al. (Eds.), *Ubiquitous intelligence and computing* (pp. 439–452). Berlin: Springer.
17. Dai, R., Wang, P., & Akyildiz, I. F. (2012). Correlation-aware QoS routing with differential coding for wireless video sensor networks. *IEEE Transactions on Multimedia, 14*(5), 1469–1479.
18. Lee, S.-K., Koh, J.-G., & Jung, C.-R. (2014). An energy-efficient QoS-aware routing algorithm for wireless multimedia sensor networks. *International Journal of Multimedia and Ubiquitous Engineering, 9*(2), 245–252.
19. Mehmet, C. V. (2010). XLP: A cross-layer protocol for efficient communication in wireless sensor networks. *IEEE Transactions on Mobile Computing, 9*(11), 1578–1591.
20. Melodia, T., Akyildiz, I.F. (2008) *Cross-Layer Quality of Service Support for UWB Wireless Multimedia Sensor Networks*. In *INFOCOM. The 27th Conference on Computer Communications. IEEE*. 2008.
21. Farooq, M.O., Kunz, T. & St-Hilaire, M. (2011) Differentiated services based congestion control algorithm for wireless multimedia sensor networks. In *Wireless days (WD) IFIP*
22. Usman, M., et al. (2018). A joint framework for QoS and QoE for video transmission over wireless multimedia sensor networks. *IEEE Transactions on Mobile Computing, 17*(4), 746–759.
23. Shah, G. A., Weifa, L., & Akan, O. B. (2012). Cross-layer framework for QoS Support in wireless multimedia sensor networks. *Multimedia, IEEE Transactions on, 14*(5), 1442–1455.
24. Yang, X., et al. (2018). CSI-based low-duty-cycle wireless multimedia sensor network for security monitoring. *Electronics Letters, 54*(5), 323–324.
25. Mekonnen, T., et al. (2018). Energy consumption analysis of edge orchestrated virtualized wireless multimedia sensor networks. *IEEE Access, 6,* 5090–5100.
26. Guo, J., Sun, L., & Wang, R. (2012). A Cross-layer and multipath based video transmission scheme for wireless multimedia sensor networks. *Journal of Networks, 7*(9), 1334–1340.
27. Almalkawi, I.T., Guerrero Zapata, M., & Al-Karaki, J.N. (2012) *A cross-layer-based clustered multipath routing with QoS-aware scheduling for wireless multimedia sensor networks.* International Journal of Distributed Sensor Networks. p. 11.
28. Aswale, S., & Ghorpade, V. R. (2017). LQEAR: link quality and energy-aware routing for wireless multimedia sensor networks. *Wireless Personal Communications, 97*(1), 1291–1304.
29. Liu, B.-H., & Jhang, J.-Y. (2014). Efficient distributed data scheduling algorithm for data aggregation in wireless sensor networks. *Computer Networks, 65,* 73–83.
30. Hermans, A., (2012) *H.264/MPEG-4 Advanced Video Coding*.
31. Jiyan, W., Xiaokun, W. & Junliang, C. (2013) SPMLD: Sub-packet based multipath load distribution for real-time multimedia *Traffic*. In *vehicular technology conference (VTC Fall), 2013 IEEE 78th*. 2013.
32. Gui, X., M.N.S., Swamy, & Ahmad, M.O. Optimal packet scheduling for multi-description multi-path video streaming over wireless networks. In *Communications, 2007. ICC '07. IEEE International Conference on*. 2007.
33. MEMSIC Technology, http://www.memsic.com/products/wireless-sensor-networks.html. Accessed on March 2012.
34. Amjad, M., Rehmani, M. H., & Mao, S. (2018). Wireless multimedia cognitive radio networks: A comprehensive survey. *IEEE Communications Surveys & Tutorials, 20*(2), 1056–1103.
35. Marano, S., Willett, P. (2018). Resource allocation in energy-harvesting sensor networks. *IEEE Transactions on Signal and Information Processing over Networks*, *4*(3), 585–598. https://doi.org/10.1109/TSIPN.2018.2800645.

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